Water Quality Prediction and Classification using Deep Learning

Abstract

Access to clean and safe water is essential for public health, environmental sustainability, and effective resource management. Accurate assessment of groundwater quality enables environmental agencies and policymakers to identify contamination risks and support informed decision-making. This project develops Deep Learning models to predict Water Quality Index (WQI) and classify groundwater quality using physicochemical parameters collected by the Central Pollution Control Board (CPCB), India.

The study utilizes a dataset containing over 19,000 groundwater samples collected across multiple Indian states between 2019 and 2022. A comprehensive analytical workflow was implemented, including exploratory data analysis, data preprocessing, baseline machine learning modeling, artificial neural network development, advanced deep learning experimentation, and hyperparameter optimization using Optuna.

Results demonstrate that groundwater quality can be predicted with exceptionally high accuracy using readily available water chemistry parameters. The baseline ANN achieved an R^2 score of 0.999984 for WQI prediction and a weighted F1 score of 0.9777 for water quality classification. In addition to model development, the study systematically evaluated several Deep Learning strategies, including optimizer selection, Batch Normalization, Dropout Regularization, logarithmic target transformation, and automated hyperparameter tuning. Despite the extensive experimentation, these advanced techniques produced only marginal or no improvements over the baseline architecture, indicating that the dataset contains highly informative predictive features and strong feature-target relationships.

The findings highlight an important machine learning principle: increasing model complexity does not necessarily improve performance when the underlying data already provides strong predictive signals. The project demonstrates both the effectiveness of Deep Learning for groundwater quality assessment and the importance of evidence-based model selection in practical machine learning applications.

1. Introduction

Groundwater serves as a major source of drinking water for millions of people. However, water quality is influenced by geological conditions, agricultural practices, industrial activities, and environmental factors. Monitoring groundwater quality is therefore essential for ensuring public health and sustainable water resource management.

Traditional water quality assessment methods require laboratory testing and expert interpretation. Machine Learning and Deep Learning provide opportunities to automate this process by learning relationships between water chemistry parameters and water quality indicators.

The objectives of this project are:

1. Predict Water Quality Index (WQI) using Deep Learning Regression.
2. Predict Water Quality Classification using Deep Learning Classification.
3. Compare Deep Learning models with traditional Machine Learning approaches.
4. Investigate the effectiveness of advanced deep learning techniques and hyperparameter optimization.

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2. Dataset Description

The dataset was obtained from the Central Pollution Control Board (CPCB), India.

Dataset Overview

- Total Records: 19,029
- Total Features: 24
- Time Period: 2019–2022

Input Features

- pH
- Electrical Conductivity (EC)
- Carbonates (CO_3)
- Bicarbonates (HCO_3)
- Chlorides (Cl)
- Sulfates (SO_4)
- Nitrates (NO_3)
- Total Hardness (TH)
- Calcium (Ca)
- Magnesium (Mg)
- Sodium (Na)

- Potassium (K)
- Fluoride (F)
- Total Dissolved Solids (TDS)

Target Variables

Regression Target

- Water Quality Index (WQI)

Classification Target

- Water Quality Classification

Classes:

- Excellent
- Good
- Poor
- Very Poor yet Drinkable
- Unsuitable for Drinking

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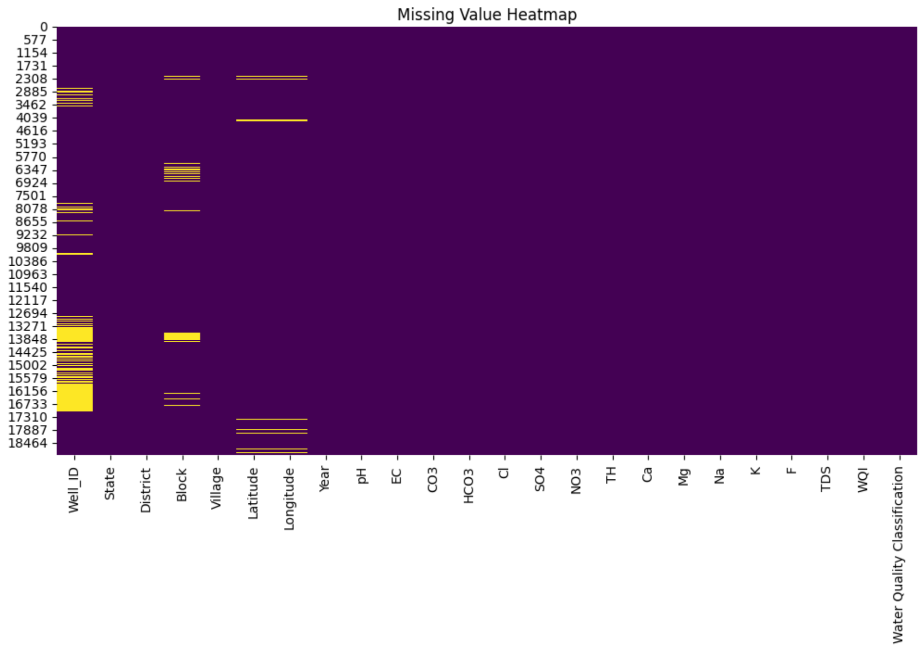
3. Exploratory Data Analysis

Dataset Quality Assessment

The dataset contained minor missing values in geographical attributes and seven duplicate records.

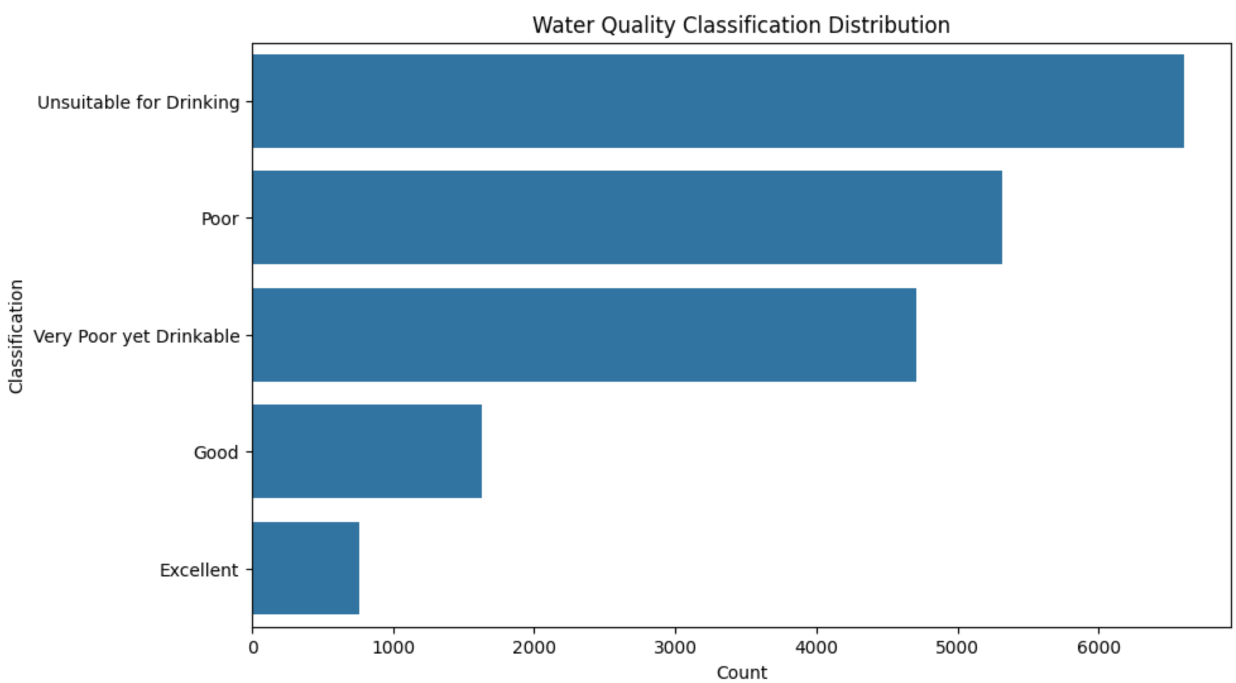
Figure 1: Missing Value Analysis



Water Quality Classification Distribution

The dataset exhibits moderate class imbalance.

Figure 2: Water Quality Class Distribution



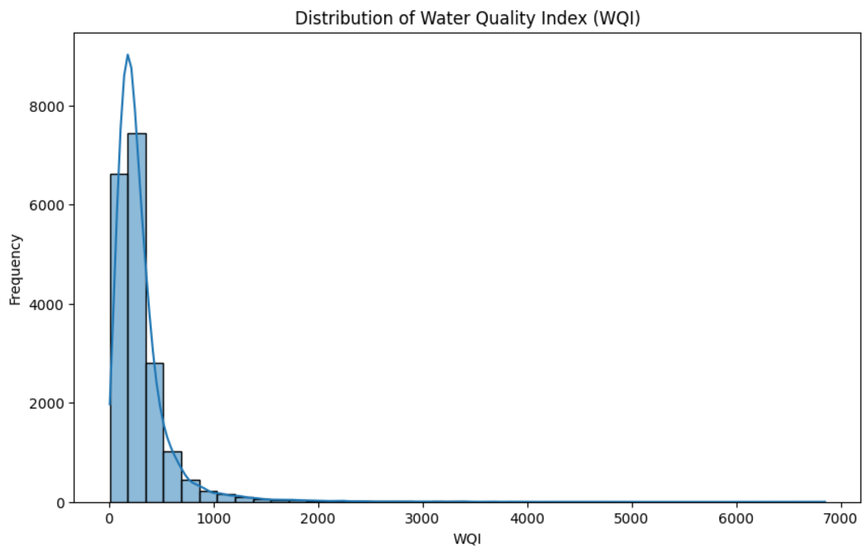
Key observations:

- Unsuitable for Drinking represents the largest class.
- Excellent quality water samples represent the smallest class.

WQI Distribution Analysis

The Water Quality Index displayed strong positive skewness due to extreme values.

Figure 4: WQI Distribution



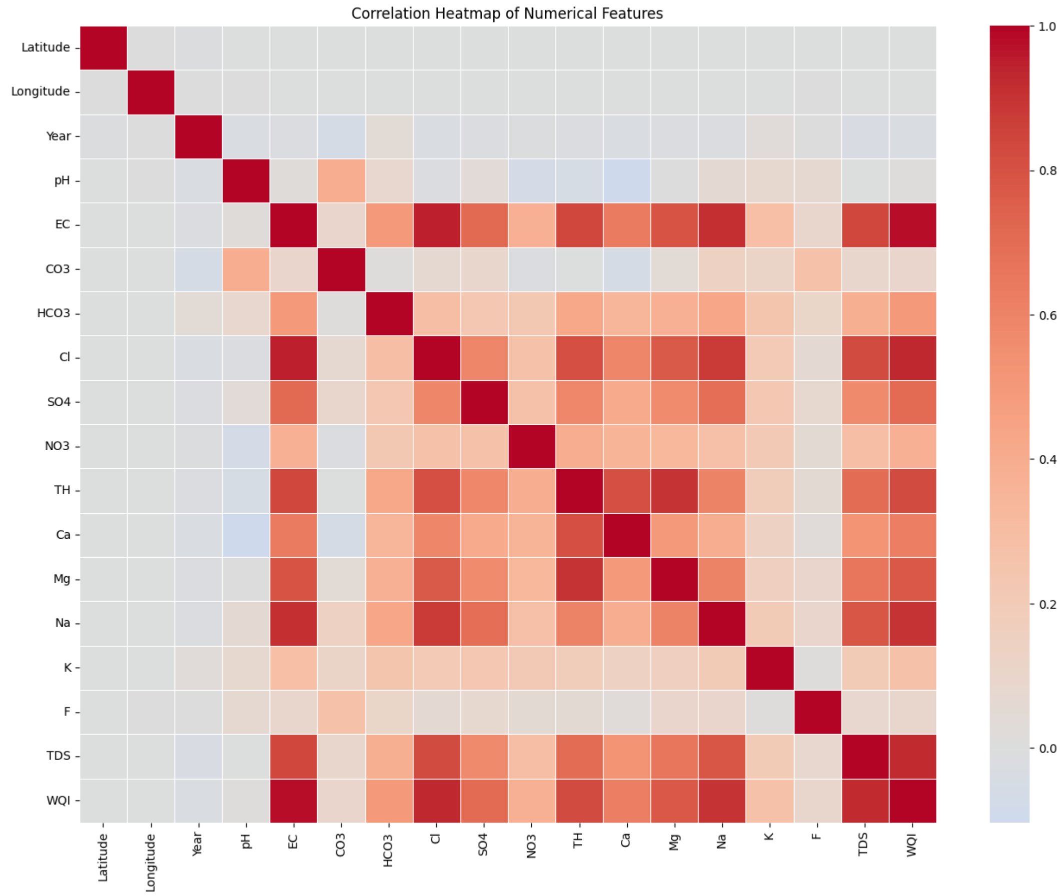
Key Statistics:

- Mean WQI: 305.29
- Maximum WQI: 6850.89
- Skewness: 6.21

Correlation Analysis

Strong correlations were observed between water chemistry variables and WQI.

Figure 4: Correlation Heatmap



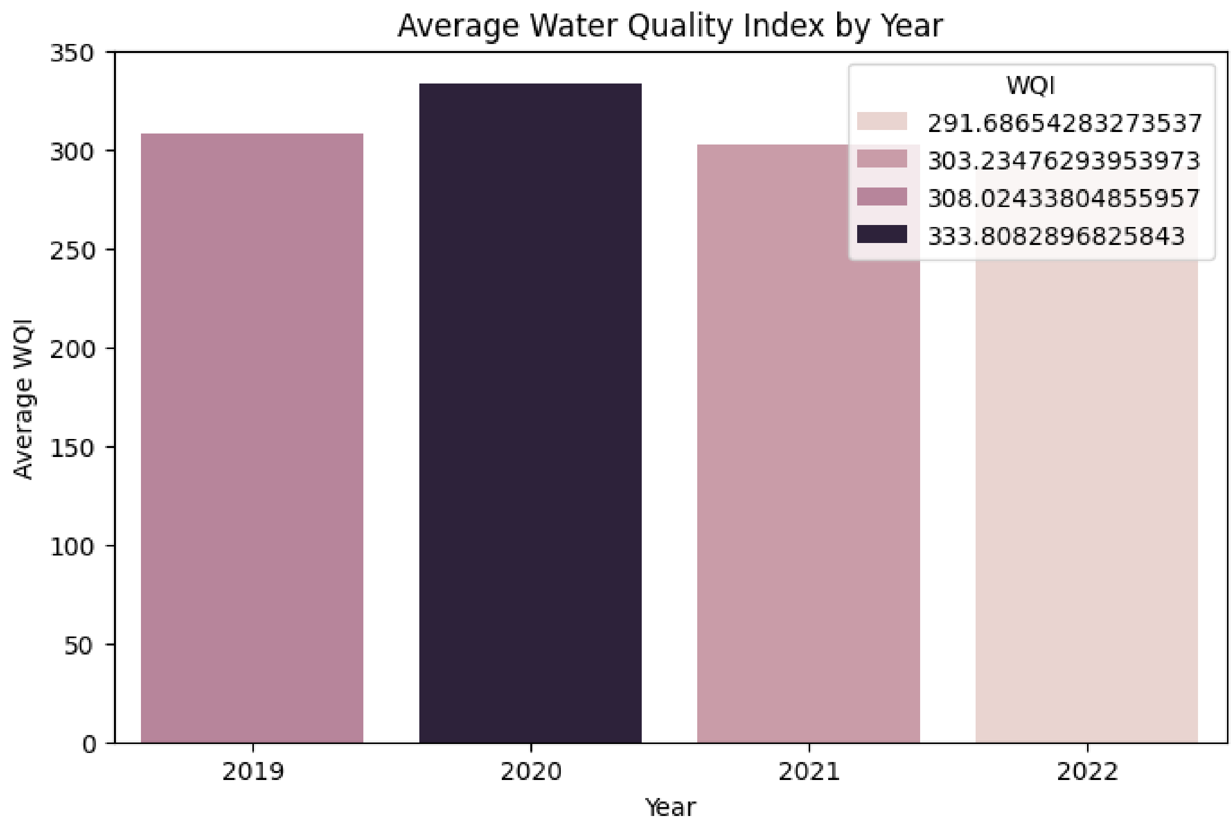
Important correlations:

Feature	Correlation with WQI
EC	0.981
Cl	0.932
TDS	0.924
Na	0.904

Temporal Analysis

Water quality trends were analyzed across years 2019–2022.

Figure 5: Water Quality Trends by Year



Key observation:

The proportion of poor-quality water remained consistently high throughout the study period.

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4. Data Preprocessing

The following preprocessing steps were performed:

- Missing value treatment
- Duplicate removal
- Label encoding of categorical variables
- Feature scaling using StandardScaler
- Train-test split (80:20)

For classification tasks, target labels were encoded using LabelEncoder.

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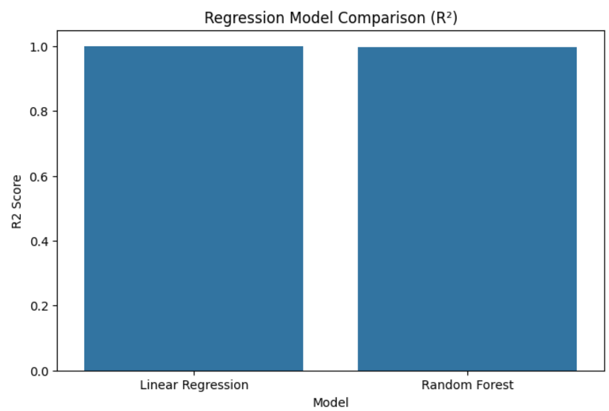
5. Baseline Machine Learning Models

Regression Models

Results

Model	RMSE	R^2 Score
Linear Regression	0.000000009	1.000000
Random Forest	18.7683	0.997030

Figure 6: Regression Model Comparison

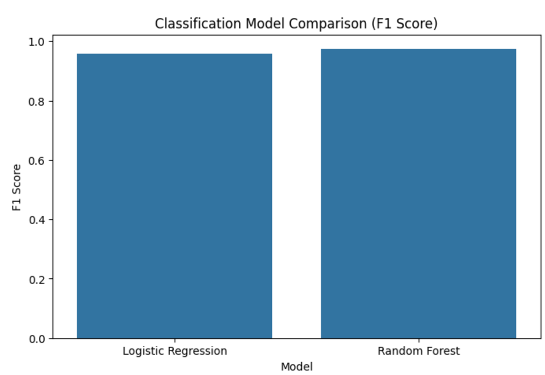


Classification Models

Results

Model	Accuracy	F1 Score
Logistic Regression	0.9585	0.9581
Random Forest	0.9740	0.9740

Figure 7: Classification Model Comparison



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6. Deep Learning Models

ANN Regression Model

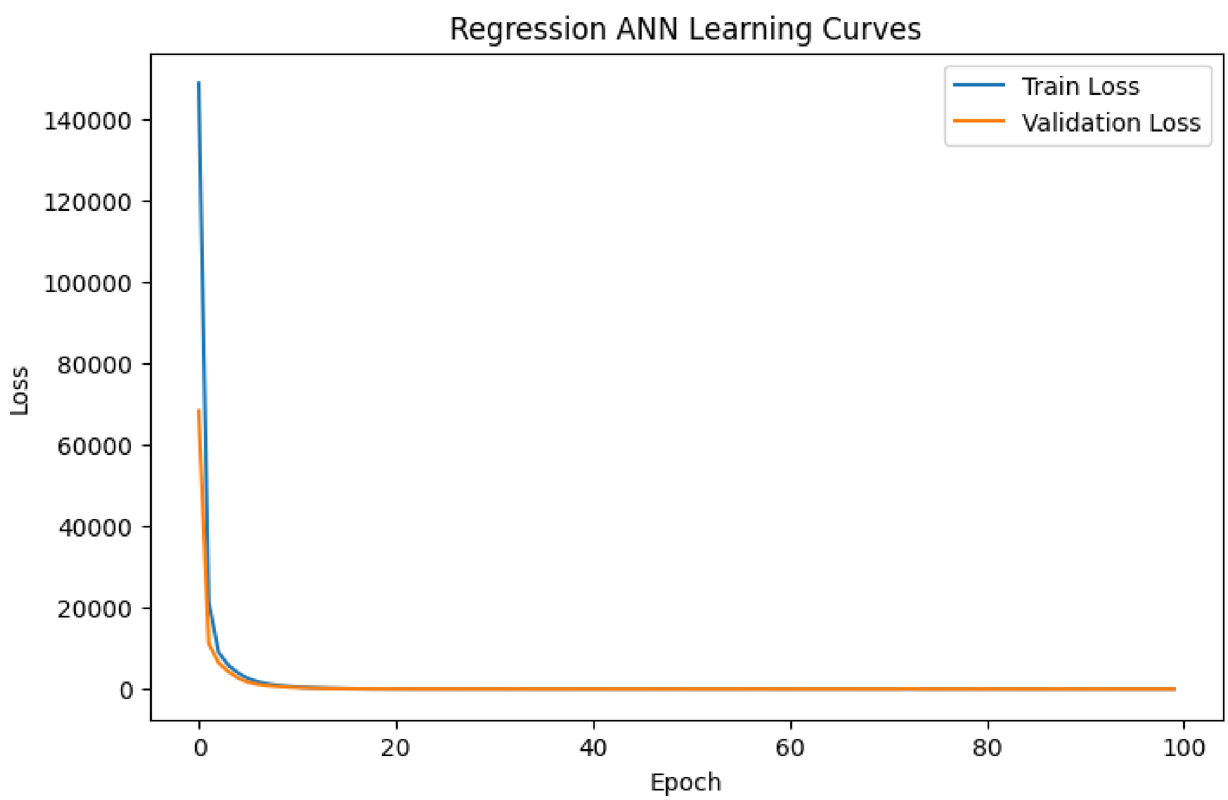
Architecture:

Input Layer → 64 Neurons → 32 Neurons → Output Layer

Results

Metric	Value
RMSE	1.3861
R^2 Score	0.999984

Figure 8: ANN Regression Training Curve



ANN Classification Model

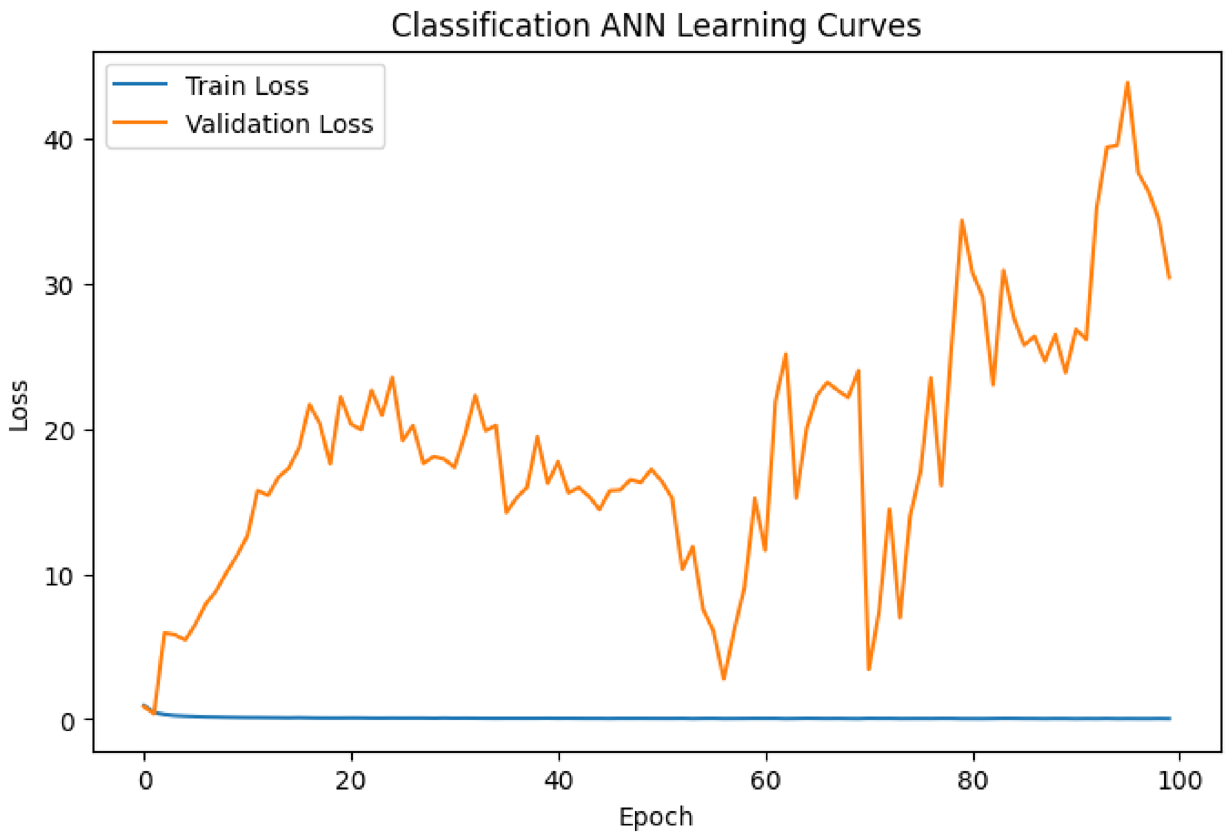
Architecture:

Input Layer → 64 Neurons → 32 Neurons → Output Layer

Results

Metric	Value
Accuracy	0.9779
F1 Score	0.9777

Figure 9: ANN Classification Training Curve



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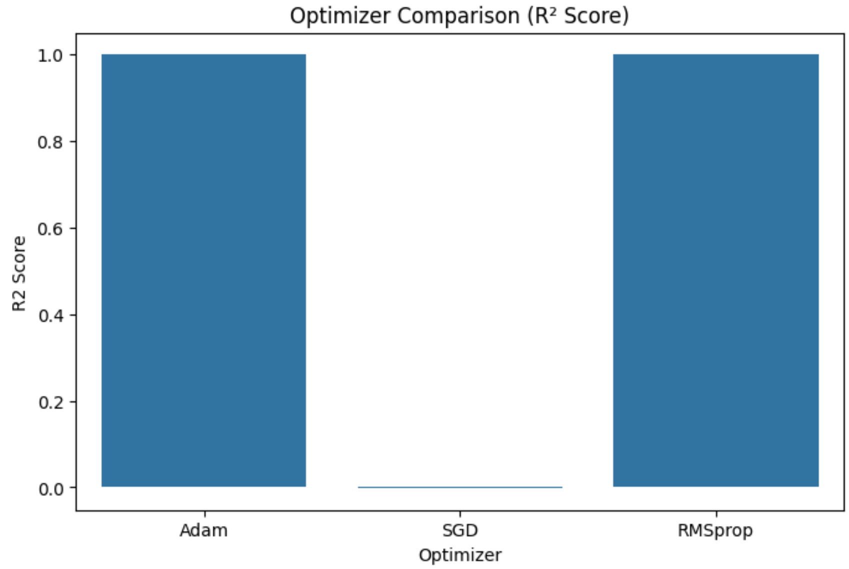
7. Advanced Deep Learning Experiments

Optimizer Comparison

Optimizers evaluated:

- Adam
- RMSprop
- SGD

Figure 10: Optimizer Comparison Results



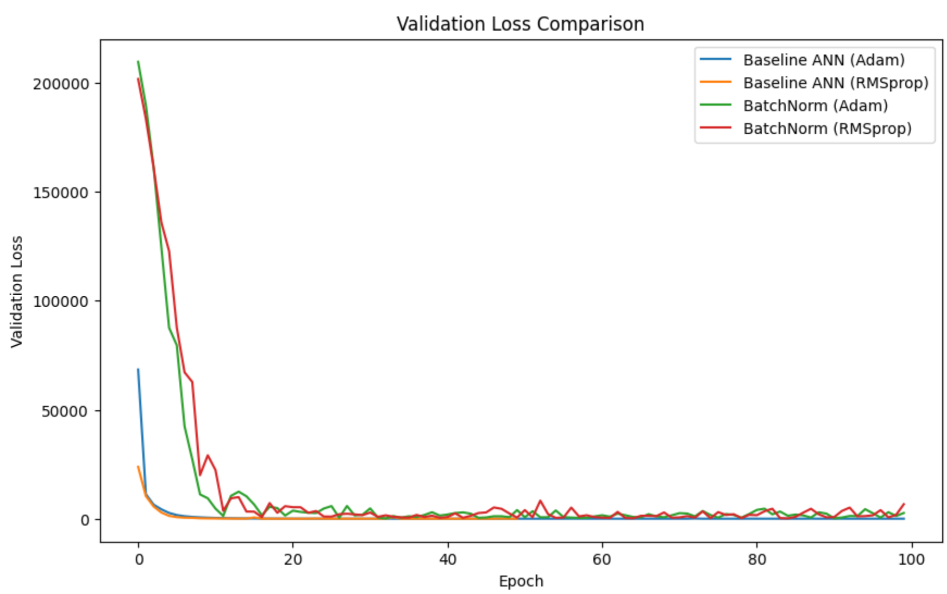
Key Finding:

Adam and RMSprop significantly outperformed SGD.

Batch Normalization

Batch Normalization was introduced after hidden layers.

Figure 10: Validation loss comparison with baseline models



Result

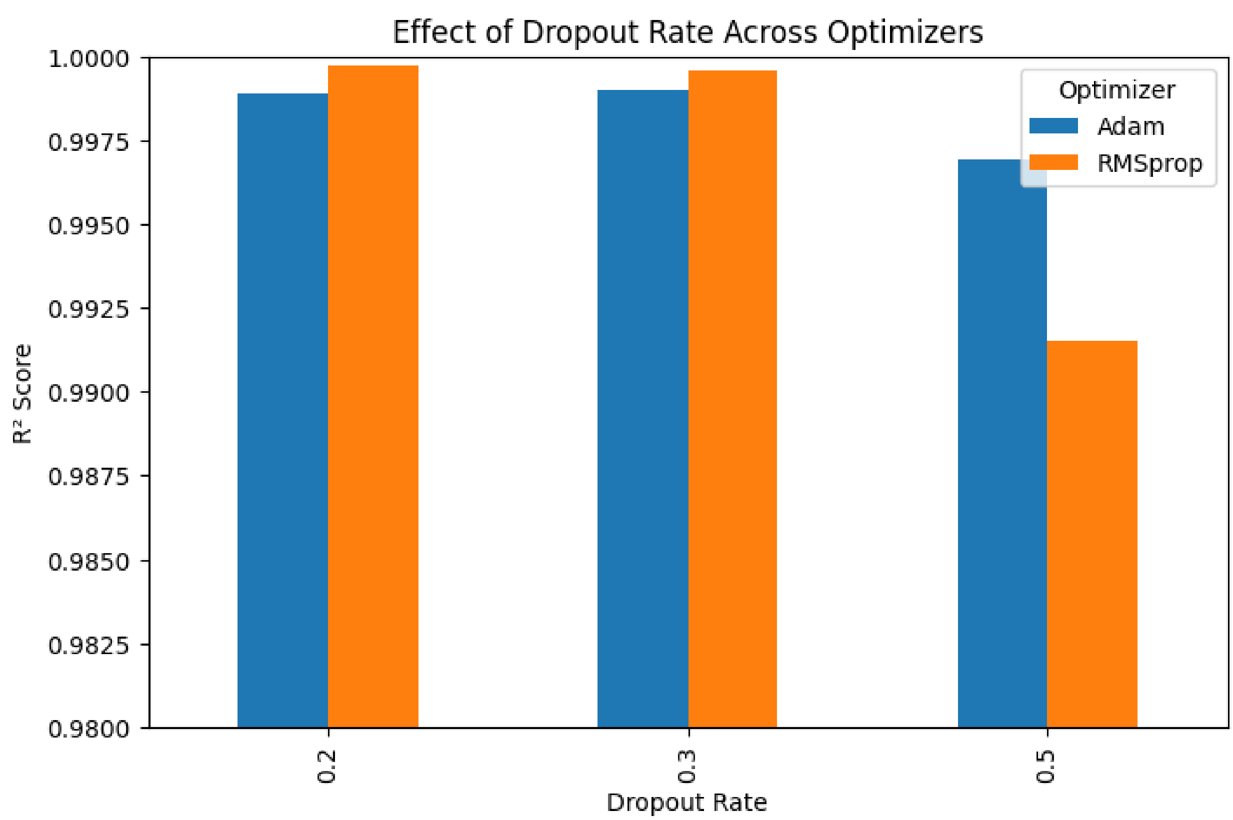
Batch Normalization reduced predictive performance compared to the baseline ANN.

Dropout Regularization

Dropout rates evaluated:

- 0.2
- 0.3
- 0.5

Figure 12: Dropout Experiment Results



Key Findings:

1. Dropout regularization was evaluated using dropout rates of 0.2, 0.3, and 0.5 under both Adam and RMSprop optimizers.
2. Moderate dropout rates (0.2–0.3) maintained high predictive performance, while aggressive regularization (0.5) significantly degraded model accuracy.
3. The optimal dropout rate varied depending on the optimizer:

- Adam performed best with Dropout = 0.3 (R^2 = 0.999005)
- RMSprop performed best with Dropout = 0.2 (R^2 = 0.999735)

4. The results reveal an interaction between optimizer selection and regularization strength, indicating that the effectiveness of dropout depends on the optimization strategy used during training.
5. Despite achieving strong performance, all dropout configurations remained inferior to their corresponding baseline ANN models without dropout.
6. Heavy regularization (Dropout = 0.5) produced the largest performance degradation for both optimizers, suggesting that excessive neuron deactivation reduces the model's ability to learn the highly structured relationships present in the dataset.
7. The relatively small difference between dropout rates of 0.2 and 0.3 suggests that only mild regularization is tolerable for this problem.
8. Overall, the groundwater quality dataset does not exhibit strong signs of overfitting, reducing the need for dropout-based regularization.

Log-Transformed WQI

A logarithmic transformation was applied to address target skewness.

Key Findings:

1. The original Water Quality Index (WQI) exhibited a highly right-skewed distribution with a skewness of 6.21, while the log-transformed WQI reduced the skewness to -0.16.

2. Log transformation significantly improved training stability and produced smoother convergence patterns for both Adam and RMSprop optimizers.

3. Training and validation loss curves became substantially less volatile compared to models trained on the original WQI values.

4. Despite improved optimization behavior, predictive performance on the original WQI scale deteriorated considerably after inverse transformation.

5. The Adam-based log-WQI model achieved an R^2 score of 0.8667, while the RMSprop-based model achieved an R^2 score of 0.9489, both significantly lower than their corresponding raw-WQI models.

6. The reduction in predictive performance suggests that the logarithmic transformation compressed extreme WQI values, causing the model to pay less attention to high-magnitude observations.

7. Small prediction errors in logarithmic space were amplified after applying the inverse exponential transformation, resulting in larger prediction errors

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8. Hyperparameter Optimization using Optuna

Regression Tuning

Best Configuration:

- Hidden Layer 1: 32
- Hidden Layer 2: 128
- Learning Rate: 0.0008
- Optimizer: Adam
- Batch Size: 32

Figure 14: Optuna Optimization History (Regression)

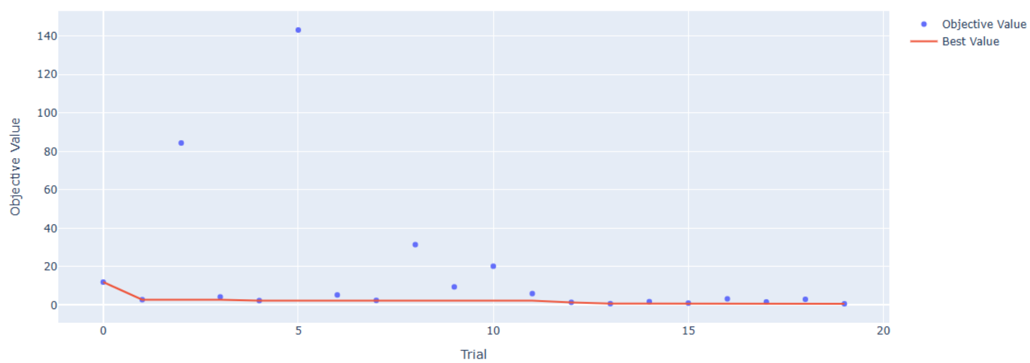
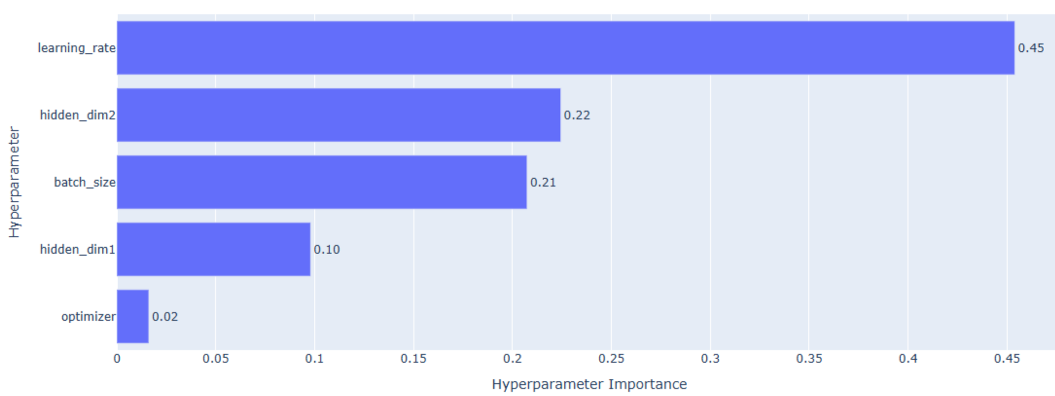


Figure 15: Hyperparameter Importance (Regression)



Classification Tuning

Best Configuration:

- Hidden Layer 1: 32
- Hidden Layer 2: 128
- Learning Rate: 0.0058
- Optimizer: RMSprop
- Batch Size: 32

Figure 16: Optuna Optimization History (Classification)

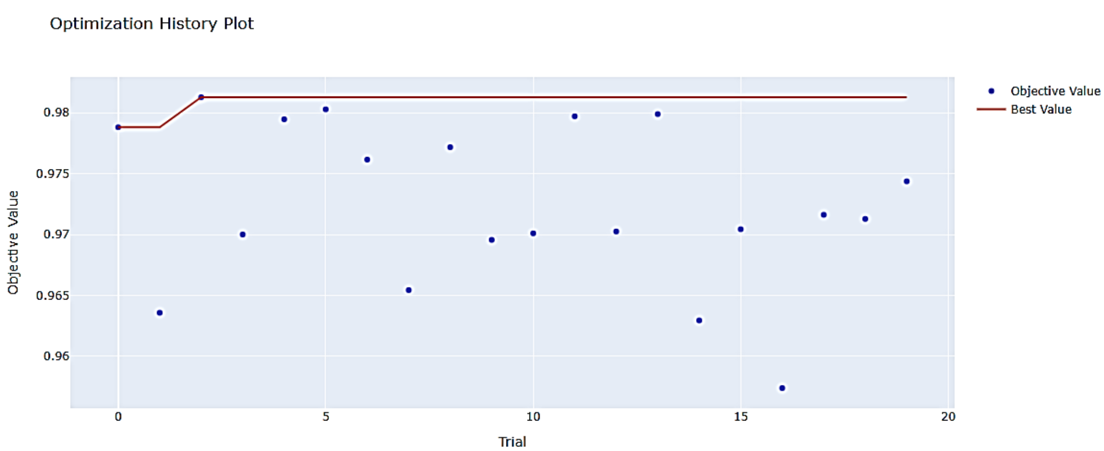
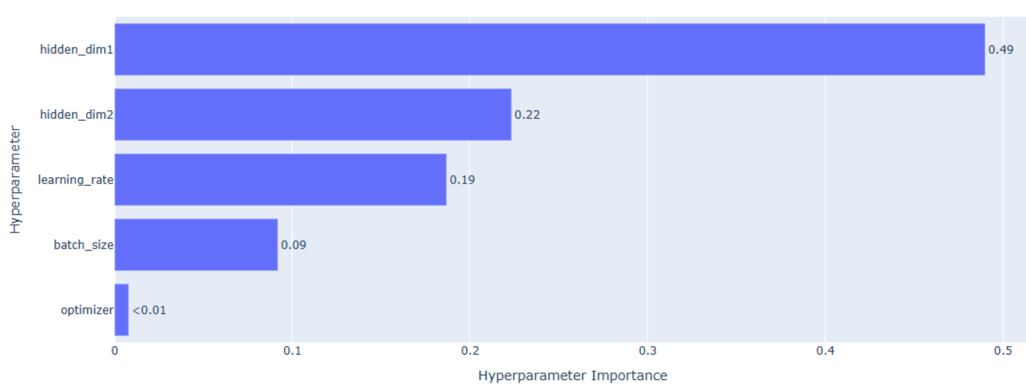


Figure 17: Hyperparameter Importance (Classification)



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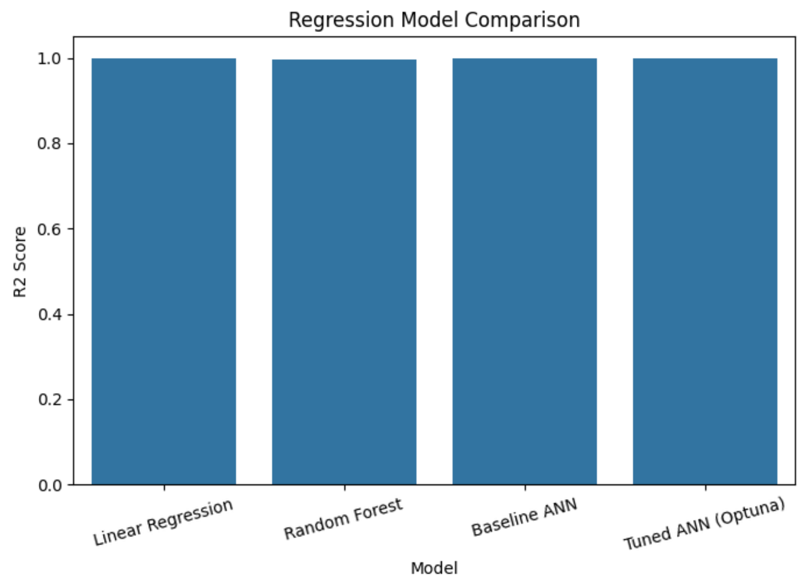
9. Final Model Comparison and Selection

Regression Models

Model	RMSE	R^2 Score
Linear Regression	0.000000009	1.000000
Random Forest	18.7683	0.997030
Baseline ANN	1.3861	0.999984
Tuned ANN	1.3245	0.999985

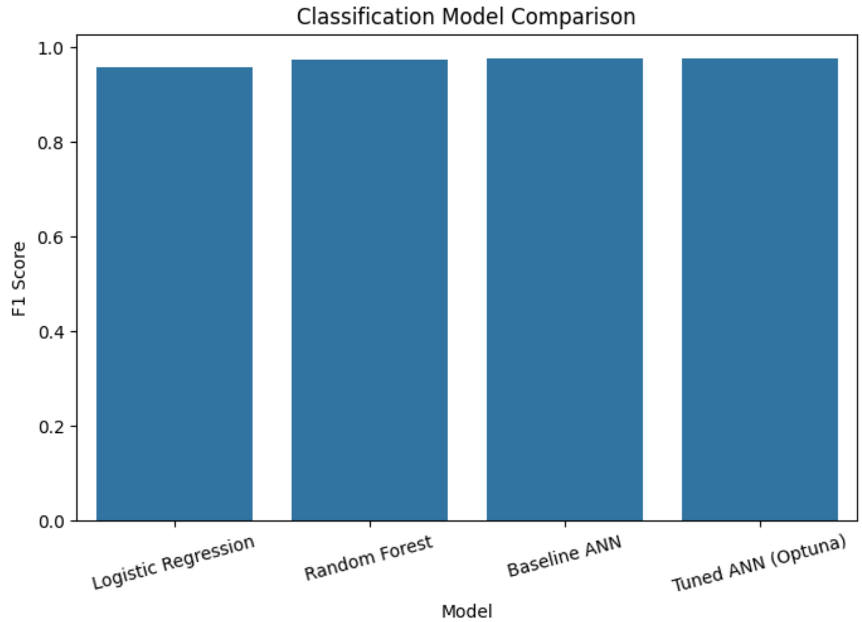
Figure 18: Final Regression Model Comparison



Classification Models

Model	Accuracy	F1 Score
Logistic Regression	0.9585	0.9581
Random Forest	0.9740	0.9740
Baseline ANN	0.9779	0.9777
Tuned ANN	0.9771	0.9770

Figure 19: Final Classification Model Comparison



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10. Conclusion

This study successfully developed and evaluated Deep Learning models for groundwater quality prediction and classification using water quality monitoring data collected by the Central Pollution Control Board (CPCB), India. The project addressed both a regression task (Water Quality Index prediction) and a multi-class classification task (Water Quality Classification), providing a comprehensive assessment of groundwater quality using physicochemical parameters.

Exploratory Data Analysis revealed strong relationships between water chemistry indicators and groundwater quality outcomes. Electrical Conductivity (EC), Total Dissolved Solids (TDS), Chlorides (Cl), Sodium (Na), and Total Hardness (TH) emerged as the most influential predictors of water quality. Correlation analysis further demonstrated an exceptionally strong relationship between EC and WQI, highlighting the highly informative nature of the dataset.

Baseline Machine Learning models achieved outstanding predictive performance, with Linear Regression producing an almost perfect fit for WQI prediction. Rather than treating these results as the endpoint of the study, extensive Deep Learning experimentation was conducted to investigate whether more sophisticated architectures and optimization strategies could provide additional benefits. Artificial Neural Networks were developed for both regression and classification tasks and achieved excellent performance, with the baseline ANN obtaining an R^2 score of 0.999984 for WQI prediction and a weighted F1 score of 0.9777 for water quality classification.

A substantial experimental effort was undertaken to systematically evaluate advanced Deep Learning techniques. Multiple optimizers (Adam, RMSprop, and SGD) were compared, Batch Normalization and Dropout Regularization were investigated, logarithmic target transformation was explored to address skewness in the WQI distribution, and hyperparameter optimization was performed using Optuna. These experiments required the development, training, evaluation, and comparison of numerous model configurations across both regression and classification tasks.

Interestingly, despite the extensive experimentation, none of the advanced techniques produced significant improvements over the baseline ANN models. Batch Normalization and Dropout Regularization generally reduced predictive performance, while Log-WQI transformation improved optimization stability but degraded prediction accuracy on the original target scale. Hyperparameter optimization identified alternative model configurations, but the gains achieved were only marginal. These findings suggest that the dataset already contains highly informative features and strong feature-target relationships, leaving limited scope for complex architectures to demonstrate additional value.

One of the key contributions of this study is the demonstration that increased model complexity does not automatically lead to better results. The project highlights an important machine learning principle: when predictive signals within the data are exceptionally strong, simpler models can perform as well as, or even better than, more sophisticated Deep Learning approaches. The value of this work therefore lies not only in the final predictive performance achieved, but also in the systematic evaluation of multiple Deep Learning strategies and the evidence-based conclusions derived from those experiments.

Overall, the study demonstrates that groundwater quality can be predicted with exceptionally high accuracy using readily available physicochemical parameters. The resulting models provide a reliable foundation for future research, environmental monitoring applications, and intelligent decision-support systems for groundwater quality assessment and management.

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11. Future Work

Future enhancements may include:

1. Time-series modeling using LSTM and Transformer architectures.
2. Integration of geospatial analysis techniques.
3. Explainable AI methods such as SHAP and LIME.

4. Ensemble learning approaches.
5. IoT-enabled real-time water quality monitoring systems.
6. Deployment as a web-based decision support system.

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12. References

1. Central Pollution Control Board (CPCB), India – Groundwater Quality Monitoring Dataset.
2. PyTorch Documentation.
3. Scikit-Learn Documentation.
4. Optuna Documentation.
5. Analytics Vidhya – Introduction to Deep Learning using PyTorch.